Monitoring Coverage Runbook: Study CP-101 (fictional)

Purpose. Keep trial monitoring running and self-escalating while the lead biostatistician is on leave. This one page tells the covering colleague (the *support*) and the study lead exactly what arrives, what it means, and who does what. It is handed over before departure and committed to version control.

> **The one honest rule.** Every email below reports that a **number crossed a pre-specified line**. That is a *prompt to look*, **not** a clinical interpretation. The covering biostatistician, the medical monitor, and the DSMB/SRC make every judgment. The system only **detects, pages, and packages**.

What runs, and when (all under service account `svc_trialmon`, not a personal login)

Job	When	Output
Data refresh + **freshness gate** + integrity	02:00 daily	confirms every feed updated; raises a flag if a source went stale
Safety scan	06:00 daily	AE/SAE · Hy's-Law/eDISH · QTcF · labs · cohort DLT tally
Routine digest → support	07:00 daily	GREEN/AMBER summary + **heartbeat**
Weekly status pack → support · one-pager → study lead	Fri 16:00	enrollment, escalation status, disposition
Threshold ALERT	on event	KRI/QTL/DLT/SAE → urgent email + escalation

What the colors mean, and who does what

- 🟢 **GREEN**: all checks within limits. **No action.** (The heartbeat still confirms the run happened on fresh data.)
- 🟡 **AMBER**: an operational / quality KRI crossed (e.g. enrollment below plan, query backlog, a deviation trend). **Support reviews and acts** (e.g. raise a site action with the PM). Stays in the routine channel; the study lead's Friday one-pager shows *flagged → being handled*.
- 🔴 **RED**: a safety threshold (DLT-defining grade, SAE, Hy's-Law lab pattern, QTcF crossing, stopping-rule count). **Support acknowledges within 30 minutes, then CONTACTS THE MEDICAL MONITOR.** The evidence packet is attached; do not wait to assemble data.

Escalation matrix

Role	Name / contact	When
Support (primary)	*[covering biostatistician]*	all emails; acts on AMBER; acknowledges RED ≤30 min
Support (backup)	*[second biostatistician / PM]*	if primary doesn't acknowledge a RED within the SLA
Role alias	`biostat-cover@…`	every RED also lands here, so it never depends on one inbox
Medical monitor	*Dr. R. Okafor, +1-…*	called by support on any RED (safety adjudication)
Project manager	*[PM]*	cc on AMBER/RED; owns operational actions
Lead (on leave)	*[contact only for true emergencies]*	reachable per the agreed emergency protocol only

If you receive **no** email (the dead-man's switch)

"No news" is **never** "all clear." A successful run always sends *something*, at minimum a GREEN digest with a heartbeat. **If the 07:00 digest does not arrive by 09:00**, the independent watcher should already have paged; if it hasn't, treat the monitoring system as **down** and notify IT/the backup immediately. A silent system is a RED condition.

Reportable vs signal-only

Everything in these emails is **signal-only** (single-program, human-confirmed): it surfaces validated outputs early. Anything that will be **formally reported** (an SRC/DSMB pack, a sponsor deliverable, a submission figure) is still **independently double-programmed and QC'd** exactly as normal before it goes out. The monitor never relaxes that standard.

Deterministic SAS/R, run inside the study's validated environment. Not executed here (no SAS license); the study team validates it as a study program. No AI, no new vendor, no PHI in email bodies. Companion to the SAS/R Trial-Monitoring Automation wiki.